The first International Conference on Electrical Engineering and Advanced Technologies, ICEEAT23
05-07 November 2023, BATNA, Algeria

Improving classification accuracy for motor imagery BCI using DWT based Features

Abdel fateh Doudou
LAS Laboratory, Electrotechnics Department, Faculty of Technology, Setif 1 University, Setif 19000, Algeria.
afdoudou19@gmail.com

Aicha Reffad
LAS Laboratory, Electrotechnics Department, Faculty of Technology, Setif 1 University, Setif 19000, Algeria.
reffad.aicha@gmail.com

Kamel Mebarkia
LIS Laboratory, Electronics Department, Faculty of Technology, Setif 1 University, Setif 19000, Algeria.
kamel.mebarkia@rwth-aachen.de

2023 International Conference on Electrical Engineering and Advanced Technology (ICEEAT) | 979-8-3503-8348-5/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICEEAT60471.2023.10426300

***Abstract*— Brain-computer interfaces (BCIs) act as vital bridges connecting human brains to external devices. They help people with disabilities do things on their own, like tasks and talking to others. One of the challenges of the BCI motor imagery electroencephalography (MI-EEG) based is the low accuracy of the imagined movements recognition. This paper introduces new features based on the discrete wavelet transform (DWT) to make MI-EEG signal decoding more accurate. Our system utilizes 18 features extracted from both the time and the frequency domains to feed a pre-trained classifier in order to identify hand movement. Many classifiers were used in this study namely the support vector machine (SVM), logistic regression (LR), k-nearest neighbors (KNN), and linear discriminant analysis (LDA). The effectiveness of the proposed system was evaluated based on the classification accuracy (CA) using the BCI Competition II Dataset III. The results of our experiments demonstrate that the LR classifier outperforms all studied classifiers achieving an impressive CA of 92.14% while using the LR classifier with just 6 features selected by genetic algorithm optimization (GA).**

***Keywords*—brain–computer interface (BCI), electroencephalogram (EEG), motor imagery (MI), features extraction, LR classifier, genetic algorithm.**

I. Introduction

Brain-computer interfaces (BCIs) serve as groundbreaking assistive technologies, acting as conduits between our brains and external devices, enabling individuals with disabilities to communicate and perform daily tasks independently [1]. As a system, it has the ability to interpret certain brain signals, including aspects of our emotions and thoughts. Generally, the brain's electrical activity is measured using an electroencephalogram (EEG). EEG is a simple and safe way that uses electrodes placed on the scalp surface to collect brain waves [2]. Brain waves are classified according to their frequency bands, and each frequency band is related to various mental states and cognitive activities. In general, there are five types of brain waves: Delta [0.5-4 Hz], Theta [4-8 Hz], Alpha [8-12 Hz], Beta [12-30 Hz], and Gamma [>30 Hz] [3].

One type of non-invasive BCI is Motor Imagery EEG (MI-EEG), which captures imagined movement patterns in order to help with control and rehabilitation [4], [5]. MI is a widely used cognitive task in which subjects are directed to mentally simulate themselves executing a particular physical action, such as moving a left hand, right hand, mouth, or foot without physically acting [6]. Each specific mental task corresponds to a distinct MI class. Many methods have been tried to evaluate MI tasks in BCIs, but EEG-based approaches are preferred because they offer an effective, high temporal resolution and low-cost device method to measure MI tasks without invasive procedures [7], [8].

Understanding this brain activity poses a challenge in EEG-based BCI systems. The MI-EEG data recorded from subjects is identified by employing a classification model. This identification process involves several key stages, including preprocessing, feature extraction, feature selection, and classification. Lately, machine learning methods have shown reliability in recognizing various MI activities [3], [6]. These methods efficiently extract features that describe pertinent information within EEG signals. After this extraction, classification techniques are used to categorize and differentiate MI activities. To achieve successful MI-EEG signal classification, both precise and swift feature extraction, as well as an effective classifier, are essential components. They play crucial roles in the classification models' effectiveness. Common feature extraction techniques encompass wavelet transform (WT) [3], Fourier transforms [9], common spatial patterns (CSP) [10], power spectral density (PSD) [11], and band power (BP) [12].

The majority of current BCIs EEG-based utilize machine learning algorithms, employing diverse classifier types. Classification methods like linear discriminant analysis (LDA) [4], [13], support vector machine (SVM) [2], [5], [14], neural network (NN) [5], k-nearest neighbor (KNN) [15], [16], Logistic regression (LR) [17], to not name all are used for BCI systems.

This work focuses on making the BCI system better by enhancing classification accuracy (CA) using MI-EEG signals. We designed a system based on Beta-band oscillations (frequency power) of EEG signals. We proposed a combination of 18 features extracted from both the time domain and frequency domain via DWT. These features are then fed to a classifier to identify left and right hand movements. Many classifiers have been investigated to choose the most efficient one.

II. Methodology

A. BCI system

Deciphering the intention to move based on brain activity patterns is a significant challenge in BCI. Fig. 1 illustrates a graphical representation that demonstrates the structure of a BCI system. Once the data has been preprocessed and the relevant features have been extracted, the intended movement of MI-EEG is accomplished using a pre-trained classifier, which has been previously trained on a dataset with relevant features to accurately predict and interpret user intentions.

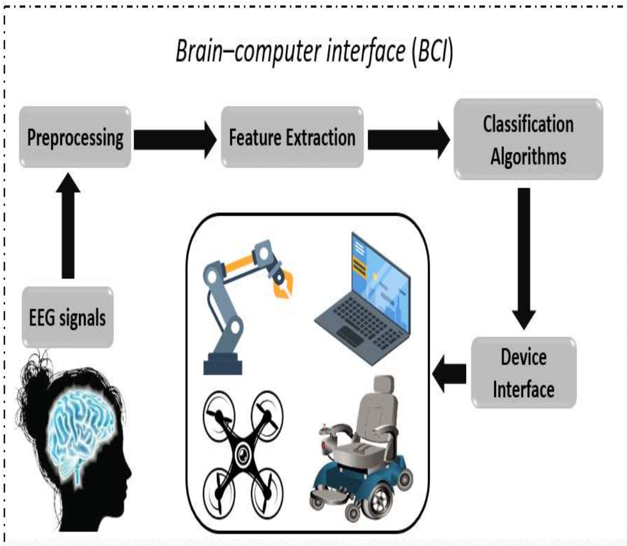

Fig. 1. A graphical representation of a BCI system.

Subsequently, a device interface can generate customized commands for specific applications.

B. Dataset description

To test the suggested approach, a publicly available BCI EEG dataset from BCI Competition II is employed to confirm its feasibility. The dataset is Dataset III of BCI Competition II (BCI Dataset III) [18]. This dataset was recorded over channels C3, Cz, and C4 from a normal subject (female, 25 years). The experiment consists of two classes MI-EEG signals that were recorded while the subject imagined movements of her left and right hand. A total of 140 trials were employed for training, and an additional 140 trials were used for testing. Each trial has a duration of 9 s. In each trial, the first 2s was quiet, at t=2 s an acoustic stimulus indicates the beginning of the trial with a fixation cross '+' displayed on the screen for 1s, and at t=3 s a visual cue (right or left arrow) is displayed. Fig. 2 and 3 illustrate the placements of the electrodes and the timing scheme measurement protocol indicating a left hand movement case. The recording utilized a G.tec amplifier along with Ag/AgCl electrodes. The dataset was sampled at 128 Hz and filtered within the range of 0.5 Hz to 30 Hz with a notch filter. Signals of channels C3 and C4 are referred to as sc3 and sc4.

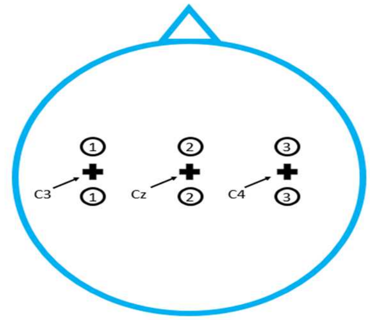

Fig. 2. The placement of electrode positions.

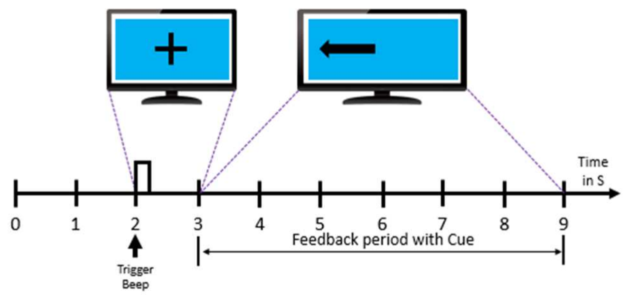

Fig. 3. Timing scheme of the BCI Competition IIDataset III.

C. Preprocessing of MI-EEG signals

Basically, the EEG signal is displayed in the time domain (TD). However, when using this domain, there are constraints on the extent of frequency and spectral details available for analysis. Thus, this study utilizes both time domain (TD) and time-frequency methods for analysis.

D. Feature extraction

Firstly, we apply a second-order Butterworth band-pass filter to the signals sc3 and sc4 within the specific range of [12 29] Hz (Beta wave) to improve accuracy and efficiency in MI tasks, remove noise, and leverage the characteristics of Beta waves. The selection of a Butterworth filter is based on its known advantages in maintaining signal integrity and minimizing phase distortion in this application, making it a suitable choice for our purposes.

In MI-EEG analysis, feature extraction plays a vital role by transforming complex EEG signals into informative representations. These extracted features capture the most important patterns and characteristics associated with the imagined brain tasks. Choosing the right features is of paramount importance, as it significantly influences the effectiveness of the CA.

In this study, we employ a diverse set of features encompassing both the time domain and the time-frequency domain (DWT) to effectively discern between right and left-hand movements. Features are extracted from the resulting filtered sc3 and sc4 signals. The statistical operators and DWT used to compute the proposed features are described below:

Kurtosis is a way to measure how data is spread out, focusing on the tails and peaks of its distribution.

$$kurt = \frac{1}{N\sigma^2}\sum_{n=1}^{N}(x_n - \mu)^4 \quad (1)$$

Where μ is the mean and σ is the standard deviation of the signal.

Root Mean Square (RMS) is a mathematical measure used to find the average value or magnitude of a set of numbers, typically a sequence of values over time, and is given by the following formula:

$$rms = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i)^2} \quad (2)$$

First difference represents the change or difference between consecutive data points in a sequence or time series and is computed with the following formula:

$$FD = \frac{1}{N-1}\sum_{i}^{N-1}|x_{i+1} - x_i| \quad (3)$$

Discrete wavelet transform (DWT) is particularly favored for time-frequency analysis because it can provide a high level of detail and resolution in the time domain while capturing frequency information. This makes it well-suited for MI-EEG signals analysis. We used the Daubechies 1 wavelet (db1) for a three-layer wavelet decomposition. Fig. 4 shows a three-layer decomposition of DWT.

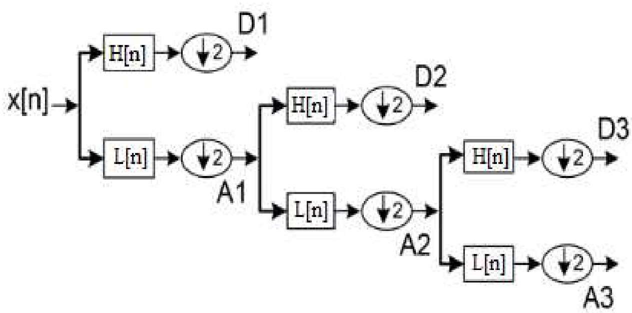

Fig. 4. DWT Wavelet Decomposition up to the third level.

The formula used to compute the DWT is as follows:

$$A[k] = \sum_n x[n].L[2k - n] \quad (4)$$

$$D[k] = \sum_n x[n].H[2k - n] \quad (5)$$

Where $x(n)$ represents an input signal, *A[k]* is the approximation coefficients, *D[k]* is the detailed coefficients, n represents the sampling point of the signal, and the functions *L[n]* and *H[n]* represent the filter coefficients.

In these formulas, 'x' represents the signal to be decomposed, and 'N' corresponds to the length of the signal (sample number). The 18 features proposed in this work are outlined in the following table:

TABLE I. PROPOSED FEATURES AND THEIR EXPRESSIONS.

Features	Expressions
f_1	$FD(sc3)/FD(sc4)$
f_2	$kurt(sc3)/kurt(sc4)$
f_3	$kurt(sc3)/var(sc4)$
f_i	$mean(A_{(i-3)4})/mean(A_{(i-3)4})$i=4…6
f_i	$rms(A_{(i-6)3})/rms(A_{(i-6)4})$i=7…9
f_i	$std(D_{(i-9)3})/std(D_{(i-9)4})$i=10…12
f_i	$kurt(D_{(i-12)3})/kurt(D_{(i-12)4})$i=13…15
f_i	$var(D_{(i-15)3})/var(D_{(i-15)4})$i=16…18

E. Machine Learning Algorithms

Once the MI-EEG signal features are calculated, we employed them to feed four well-known classifiers in MI-EEG BCI, which are the SVM, LR, KNN, and LDA classifiers to evaluate the effectiveness of the proposed features.

Logistic regression (LR) is a tool in machine learning that helps us decide between two choices, like 'yes' or 'no.' It looks at some information we have and predicts the chance of one choice being true [17]. This is really useful for tasks like deciding if an email is spam or a medical diagnosis.

Support Vector Machine (SVM) is a classification tool that separates things into different groups by finding the best separators between them [2], [10]. SVM is often used to differentiate between different types of MI-EEG signals, with the commonly used linear kernel.

K-nearest neighbor (KNN) is a simple and easy to implement non-parametric, supervised learning classifier that can be used for classification and regression [19]. This algorithm uses a classification technique known as 'nearest neighbors', which involves assigning an object to a class based on the classes of its closest neighbors [20]. KNN operates through a straightforward majority vote mechanism, where the class of an object is determined by the most prevalent class among its nearest neighbors.

Linear Discriminant Analysis (LDA) is a powerful machine-learning technique that has been successfully used for a variety of tasks, including pattern recognition, classification, and data dimension reduction [21], [22], The main concept of LDA is to find the linear combination of features that maximizes the distance between the two class means thus reducing interclass variance [23].

III. RESULTS

Using MI-EEG signals from Dataset III of BCI Competition II, the proposed features were inputs of many classifiers to increase the chance of better hand movement identification. To address the synchronization issue, features are calculated from different starting points keeping the processing duration constant and equal to 2.5 s. After having selected the best classifier, the features are selected using GA optimization to further improve the CA.

Table II shows the CA of the hand movement recognition with respect to indicated start time positions using the indicated classifiers.

TABLE II. CA OF THE HAND/RIGHT IDENTIFICATION WITH RESPECT TO THE INDICATED START TIME POSITION USING THE INDICATED CLASSIFIERS.

Time Position in seconds	Subject			
	Classification Accuracy (%)			
	SVM	LR	KNN	LDA
3.00	74.29	79.29	80.71	81.43
3.1	80.71	81.43	76.43	83.57

3.2	82.14	85.71	79.29	79.28
3.3	80.71	86.43	76.43	81.42
3.4	81.43	85	78.57	83.57
3.5	84.29	87.14	79.29	82.85
3.6	82.14	83.57	81.43	76.42
3.7	82.86	86.43	80.71	82.85
3.8	85.71	**89.29**	80.71	80
3.9	82.14	86.43	**83.57**	**85.71**
4	82.86	84.29	82.86	82.14
4.1	83.57	82.86	77.86	74.28
4.2	83.57	82.86	77.86	78.57
4.3	82.86	82.86	80	80
4.4	82.86	85.71	68.57	80
4.5	80.71	83.57	77.85	75
4.6	**86.43**	85	72.86	74.28
4.7	77.14	78.57	71.42	70
4.8	80.71	79.57	74.28	73.57
4.9	79.29	77.86	72.14	70.71
5	76.43	77.1429	67.85	67.86

Bold numbers indicate the highest CA for each classifier.

Based on these results, it's evident that LR is the most suitable choice for these applications with a CA of 89.29% with an optimal starting point equal to 3.8 s. On the other hand, the experimental findings highlight LR's superiority over KNN, LDA, and SVM classifiers.

To boost CA, the LR classifier was improved via genetic algorithms (GAs), which imitate natural selection for optimal solutions. GAs are advantageous for intricate problems and large search spaces. Using GA optimization, we reduced redundant features and minimized misclassification errors by selecting the most pertinent features.

The GA optimization uncovers the optimal combination among the 2^{18} possibilities. Fig. 5 illustrates the process of GA optimization employed to choose appropriate features. In comparison to the other 2^{18} feature sets, the features set selected by GA demonstrates superior capability in distinguishing MI-EEG resulting in improved classification results. The outcome reveals that, through the use of LR and GA optimization, the recognizer achieved a peak of 92.14% CA with only 6 features.

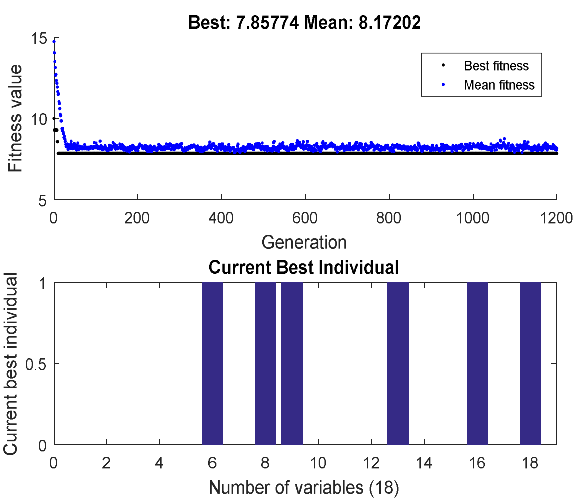

Fig. 5. GA optimization to select appropriate features for the LR classifier.

Furthermore, we delve into the accuracy of each class, with a specific emphasis on the performance of left-hand (LH) and right-hand (RH) movement tasks. Fig.6 illustrates the confusion matrix for the LR classification using selected features (SF) found by the GA. The numbers 0 and 1 on both axes correspond to the left-hand (LH) and right-hand (RH) movement classes respectively. Basing on Fig. 6, it is noticeable that the RH movement ('1' hand movement task) has been accurately identified with a rate of 97.1%, followed by LH movement with 87.1%.

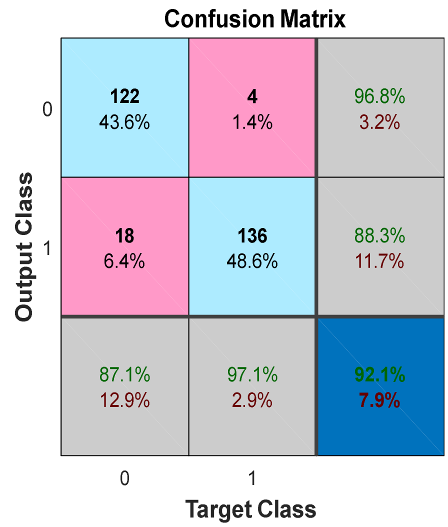

Fig. 6. Confusion matrix of the subject.

The obtained results demonstrate the effectiveness of the suggested features in classifying brain tasks via the MI-EEG signals.

Next, we will assess our method by comparing its performance with the performances of the previous methods applied to the same datasets (BCI Competition II Dataset III). This comparative analysis will provide valuable insights into the effectiveness of our techniques within the same database. In Table III, we present a summary of CA achieved by various studies, including our own, in the context of distinguishing between left and right-hand movements using MI-EEG signals. These studies utilized a variety of feature extraction techniques and classifiers to achieve their results. From Table III, it is clear that the selected features subset among the 18 proposed features, combined with LR, significantly improved the CA and reached 92.14 %.

TABLE III. CA COMPARISON OF THE PROPOSED METHOD WITH PREVIOUS WORKS.

Dataset	works	features	Classifiers	Acc (%)
	Liu et al. (2010) [10]	Common spatial pattern (CSP)	SVM	82.86
	Jang et al. (2016 [19]	STFT	KNN	83.57
Dataset	Khasnobish and	Average band power	KNN	84.29

	Bhattacharyya. (2010) [20]	of alpha and beta		
	Wang et al. (2019) [22]	TQWT	LDA	88.11
	This work	**6 Selected features**	**LR**	**92.14**

IV. CONCLUSION

The BCI system performance relies on the classifiers used to decode the brain tasks. The accuracy of classifying left/right hand movement via MI-EEG signals is significantly influenced by the quality of features and the appropriate choice of classifiers. In this paper, we have proposed 18 features that are investigated using different classifiers. Our results reveal that the LR achieved the highest classification performance. Furthermore, we enhanced this accuracy using GA optimization to identify the most informative features set for the LR classifier. By examining the visualization, it becomes apparent that GA does not only accomplish the reduction of the feature set (6 features) but also enhances CA. After comparing our approach using publicly available BCI competition databases, we achieved a maximum accuracy of 92.14% with only 6 features on the BCI Competition II Dataset III. Using only 6 features is a very important result for the BCI system as it would be faster. These results confirm that the proposed method surpasses the existing methods in classification performance offering at the same time less computational complexity by minimizing the number of features.

ACKNOWLEDGMENT

The authors would like to acknowledge the Institute for Knowledge Discovery (Laboratory of Brain-Computer Interfaces), Graz University of Technology, for providing the dataset online. (https://www.bbci.de/competition/ii).

REFERENCES

[1] Q. Yuan, W. Zhou, F. Xu, Y. Leng, and Wei D, "Epileptic EEG identification via LBP operators on wavelet coefficients," Int. J. Neural Syst., vol. 28, August 2018.

[2] K. Mebarkia and A. Reffad, "Multi optimized SVM classifiers for motor imagery left and right hand movement identification,"Australas. Phys. Eng. Sci. Med., vol. 42, no. 4, pp. 949-958, August 2019.

[3] E. Rossi, S. Pereira Soares, Y. Prystauka, M. Nakamura, and J. Rothman, "Riding the (brain) waves! Using neural oscillations to inform bilingualism research," Bilingualism: Language and Cognition, vol. 26, no. 1, pp. 202-215, 2023.

[4] D. De Massari, C.A. Ruf, A. Furdea, T. Matuz, L. Van Der Heiden, S. Halder, S. Silvoni, and N. Birbaumer, "Brain communication in the locked-in state," Brain, vol. 136, pp. 1989–2000, June 2013.

[5] F. Lotte, L. Bougrain, A. Cichocki, M. Clerc, M. Congedo, A. Rakotomamonjy, and F. Yger, "A review of classification algorithms for EEG-based brain-computer interfaces: a 10 year update," J. Neural. Eng., vol. 15, April 2018.

[6] J. Long, Y. Li, and Z. Yu, "A semi-supervised support vector machine approach for parameter setting in motor imagery-based brain computer interfaces," Cogn Neurodyn, vol. 4, pp. 207-216, 2010.

[7] S. Ge et al., "Sinusoidal signal assisted multivariate empirical mode decomposition for brain-computer interfaces," IEEE J. Biomed. Health Inform., vol. 22, no. 5, pp. 1373-1384, 2017.

[8] J. Luo, X. Gao, X. Zhu, B. Wang, N. Lu, and J. Wang, "Motor imagery EEG classification based on ensemble support vector learning," Computer Methods and Programs in Biomedicine, vol. 193, 2020, 105464.

[9] D. Coyle, G. Prasad, and T. M. McGinnity, "A time-frequency approach to feature extraction for a brain-computer interface with a comparative analysis of performance measures," EURASIP Journal on Applied Signal Processing, vol. 2005, no. 19, pp. 3141-3151, Dec. 2005.

[10] C. Liu, H.-B. Zhao, C.-S. Li, and H. Wang, "CSP/SVM-based EEG classification of imagined hand movements," Dongbei Daxue Xuebao/Journal Northeast. Univ., vol. 31, no. 8, pp. 1098-1101, 2010.

[11] C. Chao, D. Shin, A. N. Belkacem, and C. Wang, "EEG-Based Anxious States Classification Using Affective BCI-Based Closed Neurofeedback System," Journal of Medical and Biological Engineering, vol. 41, no. 1, pp. 98-108, Feb. 2021.

[12] H. Bashashati, R. K. Ward, G. E. Birch, and A. Bashashati, "Comparing Different Classifiers in Sensory Motor Brain Computer Interfaces," PLoS ONE, vol. 10, no. 6, pp. e0129435, 2015.

[13] D. Coyle, G. Prasad, and T. M. McGinnity, "A time-frequency approach to feature extraction for a brain-computer interface with a comparative analysis of performance measures," EURASIP Journal on Applied Signal Processing, vol. 2005, no. 19, pp. 3141-3151, Dec. 2005.

[14] C. Chao, D. Shin, A. N. Belkacem, and C. Wang, "EEG-Based Anxious States Classification Using Affective BCI-Based Closed Neurofeedback System," Journal of Medical and Biological Engineering, vol. 41, no. 1, pp. 98-108, Feb. 2021.

[15] M. Zhong, F. Lotte, M. Girolami, and A. Lécuyer, "Classifying EEG for Brain Computer Interfaces Using Gaussian Process," Pattern Recognition Letters, vol. 29, pp.354-359, 2008.

[16] E. Mohammadi, P. Ghaderi Daneshmand, and S. M. S. Moosavi Khorzooghi,"Electroencephalography-Based Brain-Computer Interface Motor Imagery Classification," Journal of Medical Signals & Sensors, vol. 11, no. 3, pp. 215-223, 2021.

[17] N. Md Isa, A. Amir, M. Ilyas, and M. Razalli, "Motor imagery classification in Brain computer interface (BCI) based on EEG signal by using machine learning technique," Bulletin of Electrical Engineering and Informatics, vol. 8, no. 1, pp. 269-275, 2019.

[18] B. Blankertz et al., "The BCI competition 2003: progress and perspectives in detection and discrimination of EEG single trials," in IEEE Transactions on Biomedical Engineering, vol. 51, no. 6, pp. 1044-1051, June 2004.

[19] T. U. Jang, B. M. Kim, Y. M. Yang, W. Lim and D. H. Oh, "Motor-imagery EEG signal classification using position matching and vector quantisation," International Journal of Telemedicine and Clinical Practices, vol. 1, no. 4, pp. 306-313, 2016. 39

[20] A. Khasnobish, S. Bhattacharyya, A. Konar, and D. N. Tibarewala, "K-Nearest neighbor classification of left-right limb movement using EEG data," in International Conference on Biomedical Engineering and Assistive Technologies, NIT Jalandhar, 2010. 40

[21] You, Y., Chen, W., Zhang, T, "Motor Imagery EEG Classification Based on Flexible Analytic Wavelet Transform". Biomedical Signal Processing and Control, 62, 102069, 2020.

[22] X. Wang, W. Chen, and T. Zhang, "Research of discrimination between left and right hand motor imagery EEG patterns based on tunable Q-factor wavelet transform," J. Electr. Inform. Tech., vol. 41, no. 3, pp. 530-536, 2019.